MEETING OR CONTACT REPORT

	Date of Report: July 9, 1958
Organization & Location: Los Alamos Scientific Laboratory IBM, Poughkeepsie, New York	**Date:** June 30-July 1, 1958
	Reported By: H. G. Kolsky ← D. W. Sweeney
Project: Stretch Mathematical Planning Committee Meeting #12	**Department:** 749
	Follow-up Date:

PERSONNEL PARTICIPATING:
(Place asterisk next to those on distribution list. Other distribution show at end of report)

IBM		LASL
T. Appel	*	B. Carlson *
G. A. Blaauw	*	R. Lazarus
E. Bloch	*	E. A. Voorhees
F. P. Brooks	*	M. B. Wells
W. Buchholz	*	D. Wood
E. F. Codd	*	J. Worlton
S. W. Dunwell	*	
G. A. Grover	*	
W. P. Heising	*	
F. E. Johnston	*	
H. G. Kolsky	*	
R. E. Merwin	*	
D. W. Sweeney	*	

I. Progress Report and Present Status of SIGMA.

A. General

Mr. Dunwell opened the meeting, then he and Mr. Bloch reviewed the status of the computer. Mr. Dunwell pointed out that the Stretch group would do the detailed layout and the final assembly with all of the components being manufactured in the plant.

He spoke briefly of the transistor production indicating that the PNP yields were very favorable, but that the NPN, which was started six months later, has not reached the same rate of production. Our rate of progress for the next few months will be essentially the rate of progress in producing NPNs. A third transistor, the memory driver transistor, which is an NPN, was started last and is still in pilot production, but would be in the plant by the end of the year. The quantities of these required this year are small, therefore, we are getting a sufficient number from the pilot line.

B. Memories

Mr. Dunwell then briefly spoke of the memory production. The planes for the 2 usec memory are being fabricated in the plant. They are presently being strung by hand, but automatic machinery is under development. Several planes are on hand and we expect to have the first memory in operation this year. He pointed out that when the size was doubled from 8,192 to 16,384 words, some slow-down in memory speed was expected. He stated that the speed would be 2.25 microseconds or faster. With respect to the fast memory, he stated that they expect to learn a great deal from the production of the first main memory, and that we will not start construction of the first fast memory until January 1959, although a data flow model will be made later this year. Again, because of the doubling in size from 512 to 1024 words, the speed is now .75 microseconds or faster. Mr. Voorhees asked how much of the slow-down was due to the size increase. Mr. Dunwell stated that this was an unknown factor at this time and could be determined only after final assembly.

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Dr. Wood stated that the manual intimated that we had a very fast core array in place of the 16 transistor registers specified in the contract. Mr. Dunwell stated that we were not sure how fast this core array was, but it would not be as fast as 0.2 microseconds. Mr. Bloch stated that the target speed for this core array was to produce one fast enough so as not to slow down or limit the Indexing Arithmetic Unit. There are two alternatives for the index core memories which will be investigated. One of these is a destructive read, two dimensional memory which will require 0.4 usecs for reading and 0.4 usecs for regeneration or writing. The other alternative would be a non-destructive core array with a faster read and slower destroy and write cycle (0.2 usecs read; 0.6 usec write).

Dr. Wood then asked if cores are not fast enough, would we use transistors as a backup, to which Mr. Dunwell replied, "yes". There was also discussion of the fact that the index memory could no longer hold instructions.

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Mr. Dunwell and Mr. Merwin then gave a summary of the circuits and pluggable units which have been developed. They pointed out that there are about 12 types of circuits packaged in about 24 varieties of pluggable units. These are current switching circuits requiring equal numbers of NPN and PNP transistors, but no diodes nor condensers. Two new circuits, a three-level cascode circuit and an emitter-follower circuit are also being used.

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E. The Main Frame

Mr. Bloch presented the following information about the main frame of the computer. He stated that the computer, consisting of the "I Box", look-ahead, and arithmetic unit, contained about 140,000 transistors which were roughly divided as follows:

Floating Point	25%	mostly data handling.
VFL	15%	mostly controls.
"I Box"	17%	
Look-ahead	17%	mostly controls.
Checking	26%	

He also stated that there are approximately 1000 register positions and 310 adder positions. A detailed description of these three portions of the computer is as follows:

1. Indexing Arithmetic Unit ("I Box")

The I Box has a basic internal cycle time of 0.4 usec and performs the following functions:

Accesses instruction,
Modifies instructions,
Executes index arithmetic instructions,
Modifies memory addresses,
Address compares,
Initiates all memory references,
and Transfers updated instructions to look-ahead for execution.

It will be able to prepare indexed floating point instructions at a rate of 0.8 usecs per instruction (0.4 usec is fastest rate).

2. Look-Ahead

The look-ahead will consist of four levels of information which is optimized for sequences of floating point instructions. It will contain all information about the instruction, certain recovery data and the data word associated with the instruction. It performs the following functions:

Holds all instructions to be executed by the arithmetic unit,
Fetches operands,
Performs all store operations,
and Holds all pseudo stores.

Each level of the look-ahead contains the following information:

Instruction counter,
Unmodified data field,
Operand address,
Updated indicator field,
Op-code,
and Operand.

The cycle time in the look-ahead is 0.2 usecs. The "I Box" has two full word buffer registers which hold instructions and act like a small look-ahead for the "I Box".

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The arithmetic unit has no connection to the computer except through the look-ahead. It consists of two portions:

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Executes all connect operations.
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The basic cycle of the VFL unit is something less than 0.45 usecs but the speeds in the following table are given for a 0.45 usec cycle.

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All data transfers are checked as follows:

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VFL arithmetic -- duplicated
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It was pointed out that check and transfer take about 0.35 usecs per cycle and if there were no checking the time-saving would be about 0.15 usecs.

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9. The operation Store Rounded should be defined as normalized and then round, in contrast to its present definition.

10. Los Alamos asked for some provisions to test whether the machine was in an enabled or disabled state. If this is not provided, restoration of exact machine status is difficult.

II. Programming

A subcommittee consisting of Messrs. Carlson, Wells, Voorhees, and Worlton of LASL, and Messrs. Apple, Grover, Heising, Codd, Kolsky, Brooks, and Herwitz of IBM, met separately to discuss programming plans and conventions.

Mr. Grover presented a proposal for a SAP-like coding form for STRAP (Stretch Assembly Program). Although the first STRAP code will be written for the 704, the coding formats and programming conventions should not be restricted to conform to the 704 in any fundamental way. The goal is to have an assembly program which can be expanded later and still accept codes which were written for the less advanced system. Mr. Grover proposed a coding format in which the instruction fields were separated by punctuation, i.e., blanks and commas. Mr. Worlton discussed a format based on a combination of vertical lines (absolute positioning) and punctuation.

There were several hours of discussion of these and related coding conventions. Everyone was in agreement with the general aims and philosophy of the program, although there were differences on details.

It was agreed that Mr. Worlton and Mr. Grover would try to reach an agreement by August 15, 1958, on the following STRAP conventions:

1. Coding form.
2. Card Forms (symbolic and binary).
3. Mnemonics for op codes.
4. Principal pseudo-ops to be included.
5. Output listing format.

To avoid duplication of effort, the intent is for LASL to write the STRAP code for the 704 and Applied Programming to write the supervisory system.

There was also some discussion of a 704 numerical and logical simulator for STRETCH programs. Its main purpose would be to debug programs before STRETCH was available. Decisions concerning it were deferred until more is known about STRAP.

III. Next Meeting

The date for the next meeting of the Mathematical Planning group was tentatively set for early October 1958, at Los Alamos.

DWS:HGK/pkb

cc: Mr. D. W. Pendery
Mr. A. W. Kleinebecker

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Mr. Grover presented a proposal for a SAP-like coding form for STRAP (Stretch Assembly Program). Although the first STRAP code will be written for the 704, the coding formats and programming conventions should not be restricted to conform to the 704 in any fundamental way. The goal is to have an assembly program which can be expanded later and still accept codes which were written for the less advanced system. Mr. Grover proposed a coding format in which the instruction fields were separated by punctuation, i.e., blanks and commas. Mr. Worlton discussed a format based on a combination of vertical lines (absolute positioning) and punctuation.

There were several hours of discussion of these and related coding conventions. Everyone was in agreement with the general aims and philosophy of the program, although there were differences on details.

It was agreed that Mr. Worlton and Mr. Grover would try to reach an agreement by August 15, 1958, on the following STRAP conventions:

1. Coding form.
2. Card Forms (symbolic and binary).
3. Mnemonics for op codes.
4. Principal pseudo-ops to be included.
5. Output listing format.

To avoid duplication of effort, the intent is for LASL to write the STRAP code for the 704 and Applied Programming to write the supervisory system.

There was also some discussion of a 704 numerical and logical simulator for STRETCH programs. Its main purpose would be to debug programs before STRETCH was available. Decisions concerning it were deferred until more is known about STRAP.

III. Next Meeting

The date for the next meeting of the Mathematical Planning group was tentatively set for early October 1958, at Los Alamos.

DWS:HGK/pkb

cc: Mr. D. W. Pendery
Mr. A. W. Kleinebecker